MERI COLLEGE

NAME - EKTA

COURSE - BCA

ROLL NO - 24CET0181

SUBJECT - Software Engineering

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Assignment-1

Ques1 Explain the charcteristics of Softwere and discuss the evolving role of Softwere in modern applications.

Ans: Softwere is a collection of programs procedures, and related data date instructs a computer to perform specific tacks. Unlike hardwore, Softwere is intangible and Canbe modified updated easily.

1. Intangible – Softwere Cannot be physically touched. it exists as programs and data.

2. Developed, not manufactured – Softwere is Created through analysis disgn. Coding, testing, and maintenance rather than being mantenance rather than being manufactured like hardwa

3. Does not wear out physically – Softwore does not deterioorate with use, although it may became outdated or develop faults.

4. Easy to modify – Softwere Can be updated or changed to meet new requirements.

5. Complexity – Large Sofwere systems. Can Contain millions of lines of Code and involve many interconnected components.

6. High maintenance requirement - Software requires Continou maintaence, such as bug fixing Security updates, and performance improvements.

7. Reusable - Components, Libraries, and modules can often be reused in different applications.

8. use - dependent - Software is generally disigned to meet Specific user or business requirements.

Evolving Role of Software in modern applications

the role of Software has changed significantly from simple Computational programs to intelligent and highly Connected Systems.

Business applications: Software Supports accounting, banking inventory Coustomer managment, and business automation.

Web and mobile applications: application provide Services such as shopping, Communcation, education entertainment, and onlion banking.

Cloud Computing: Software Can now be delivered though the internet, allowing users to acess applications and data from almost anywere.

Q.2. Explain the Software Crisis and discuss the need for a Software engineering approach.

Ans The Software Crisis refers to the difficulties faced in developping large and Complex Software Systems reliably on time, and within budget. It became particularly noticeable as Software Systems grew large and more Complicated.

1. Cost overruns – Software projects often cost more than orginally estimated.

2. Late dilivery - Projects may take much longer than planned.

3. Poor quality – Software can contain many errors and bugs.

4. Diffcult maintennce – modifying or updating poorly disigned Software becomes expensive and time-consuming.

5. Unreliable Software – Systems may fail or produce incorret results.

Need for a Software engineering approch –

- Software engineering applies Systematic discilpned, and measurable method to Software development. It helps Overcome the problems associated with the Software Crisis

- Better planning: Requirements, resources, time, and Costs. Can be planned Systematically.

- Improved quality: Standards, testing, reviews, and quality-Control techniques rduce errors.
- Cost effectiveness: proper development methods help redue unnecessary development and maintenance costs.
- Timely delivery: project manegment and structured development processes help meet deadlines.
- Easy maintenance: Good design and documentation makes Software easier to modify and maintain.
- Reliability and Security: Good design and documentation make Software easier to modify and maintain..
- Team Coordination: Defined processes and documentation improve Communication among developers and Stakeholders.

Q.3 Compare Waterfall, Prototype Iterative Enhancement Spiral and RAD models.

Ans- Software development models provide a systematic Approach for developing software. they define the different phases involved in software development such as requirment analysis, disign, Coding, testing, implementation, and maintenance. the major software development models are Waterfall, Prototype, Iterative, Enhancement, Spiral and RAD models.

1. Waterfall models

the Waterfall models is a traditinal software development model in wich development progresses though a series of sequential phases. Each phase must be Completed before the next phase begins.

Phases:

1. Requirement Analysis
2. System Design
3. Implementation/Coding
4. testing
5. Development
6. maintenance

Advantages:

- Simple and easy to understand.
- Proper documentation is maintained.
- Suitable when requirements are fixed.

2. Prototype model

the prototype model involves Creating an early working Version, or Prototype, of the Software. the prototype is Shown to the users to Collect feedback and understand their Requirement more early.

Process:

Requirement → Prototype → user
feedback → Refinement → final ~~Sto~~ System

Advantage.

- Help understend unclear requirment
- Provides early user feedback.

3. Iterative model

the Iterative model develops Software though reapeted cycles Called iterations Each iteration produces and improved Version of the Software.

for example – the first iteration may provide basic feature while later iterations add additinal functionality and Improvements.

Advantage

- Working Software is available early.
- problems Can be identified earlier.
- Changes can be incorporated during possible.

4. Enhancement model.

The Enhancement model develops a basic Version of Software first and than gradually adds new features and Improvements. Each new Version enhances the previous Version.

Example:

- Version 1: basic functions
- Version 2: Additinal features
- Version 3: Improved performance and functinality.

Advantage:

- Software can be delivered in smaller parts.
- Users can start using the basic System early.
- Easier to manage large Systems.

5. Spiral model.

the Spiral model combines iterative development with Systematic risk analysis Development takes place though a series of cycle or, Spirals. Each cycle focuses on planning, risk analysis development, and evaluation.

main Activities:

- Planing
- Risk Analysis
- Engeneering / Development
- Costomer Evalution.

Advantage

- Strong focus on risk mangement.
- Suitabe for large and Complex projects.

Q.4 Explain major Software project mangment, activities Including project planning and Scheduling.

Ans. Software project manegment (SPM) involves planning, Organizing, monitoring and Controlling a Software project So that it is Completed on time, within budget, and according to required quality Standards.

1. project planning

Project planning defines what needs to be done and how it will be achievid. major activities include.

- Defining project objectives and Scope
- Identifying required resources
- Estimating Cost and effort
- identifying project risk
- Selecting development methods and tools.
- preparing the project plan.

2. Project Schedulling

Scheduling determines when each project activity will be performed. it includes.

- Breaking the project into Smaller tasks
- identifying task dependencies
- Estimating the time required to each task.
- Assigning task to team members.
- Setting milestones and deadlines
- Preparing Schedules using tools Such as Gantt Charts or network diagrams.

3. Resource manegment

Resource such as developers, designers. hardware, Software and budget are identified and allocated effectively.

4. Risk manegment

Possible risks are identified, analyzed and controlled. Staff Changes, cost overruns. and Schedule delays.

5. Project monitoring and control

Project progress is regularly Compared with the planned Schdule, Cost, and Scope, Corrective actions are taken when deviations accur.

6. Quality mangment.

Standards and procedurs. are established to ensure that the Software meets Coustoner requirement and Quality Expectations.

7. Configuration and change mangment

Changes to requirment, Software Components, and documents are Controlled to prevent confusion and maintain project stability.

8. Communication and reporting

Regular, meetings, progress. reports. and documentation help keep Customers managers, and team members Informed.

9. Project Closure.

After Successful Completion, the Software is delivered, final documentation is prepared resorces are released, and lessons learned are recorded.

Conclusion.

Effective Software project managment ensures systmatic Planning, Scheduling, resorces. allocasion, risk. Control, Quality assurance. monitoring and sucessful. completion of the Software. project.

Q.5 Discuss Software risk management and explain the major activities involved in risk identification and mitigation.

Ans Software Risk mangment is the process of identifying analyzing, planning, and Controlling risks that may affect a Software project. A risk is an uncertain event that can negetively affect the cost Schdule, quality, Scope or Perfomance of the project.

1. Risk identification
identification possible risks that may occur during the Project. Common risks include.

- Requirement changes
- Lack of Skilled developers
- technology failure
- Cost overruns
- Schdules delays
- Security problems
- Hardware of Software failures

2. Risk Analysis and Assessment
Each identified risk is analyzed according to

- Probability — likelihood that the risk will occur.
- Impact — amount of damage it may Cause.

Risk Can then be classified as low, medium, or high priority.

3. Risk prioritization

Risk with high probability and high impact are given the greatest attention are given attention. this help. Project team use its limited resources effectively.

4. Risk mitigation planning

A strategy is developed to reduce the probability or impact of each imported risk. for example. if a key developer may leave the project. knowledge can be shared among mutiple team members.

5. Risk monitoring and control

Risk are continuously monitored throughout the project. new risks are identified. existing risk are reviewed and mitigation plans are updated when necessary.

6. Risk Response

the team may repond to a risk by:

- Avoiding the risk
- Reducing / Mitigation its probabality or impact.
- Transferring it to another party.
- Accepting it when the cost of prevention is to high.